

Ex. No. : 01

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GUI Components

Aim

Develop an application to change the font and color of the text and display toast message when the user presses the button.

Procedure:

Step 1 : File -> NewProject

Provide the application name and Click "Next"

Step 2 : Select the target android devices

Select the minimum SDK to run the application. Click "Next".

Step 3 : Choose the activity for the application (By default choose "Blank Activity").

Click "Next".

Step 4 : Enter activity name and click "Finish".

Step 5 : Edit the program.

Step 6 : Run the application, 2-ways to run the application.

1. Running through emulator

2. Running through mobile device

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.GUIComponents"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/rootLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <TextView
        android:id="@+id/sampleText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello, Kotlin!"
        android:textSize="18sp"
        android:textColor="#000000"
        android:layout_marginBottom="24dp" />

    <Button
        android:id="@+id/btnIncreaseSize"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Increase Text Size" />

    <Button
        android:id="@+id/btnChangeTextColor"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Change Text Color"
        android:layout_marginTop="16dp" />

    <Button
        android:id="@+id/btnChangeBgColor"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Change Background Color"
        android:layout_marginTop="16dp" />

</LinearLayout>
```

MainActivity.kt

```
package com.example.guicomponents

import android.graphics.Color
import android.os.Bundle
import android.widget.Button
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
import kotlin.random.Random
import android.widget.TextView
import android.widget.LinearLayout

class MainActivity : AppCompatActivity() {

    private var currentTextSize = 18f
    private lateinit var sampleText: TextView
    private lateinit var rootLayout: LinearLayout

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        sampleText = findViewById(R.id.sampleText)
        rootLayout = findViewById(R.id.rootLayout)

        val btnIncreaseSize: Button = findViewById(R.id.btnIncreaseSize)
        val btnChangeTextColor: Button = findViewById(R.id.btnChangeTextColor)
        val btnChangeBgColor: Button = findViewById(R.id.btnChangeBgColor)

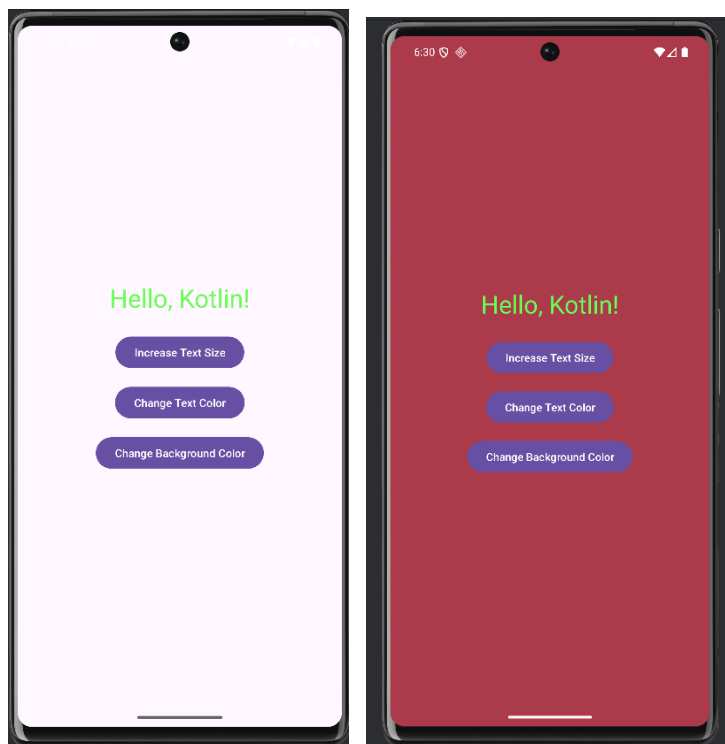
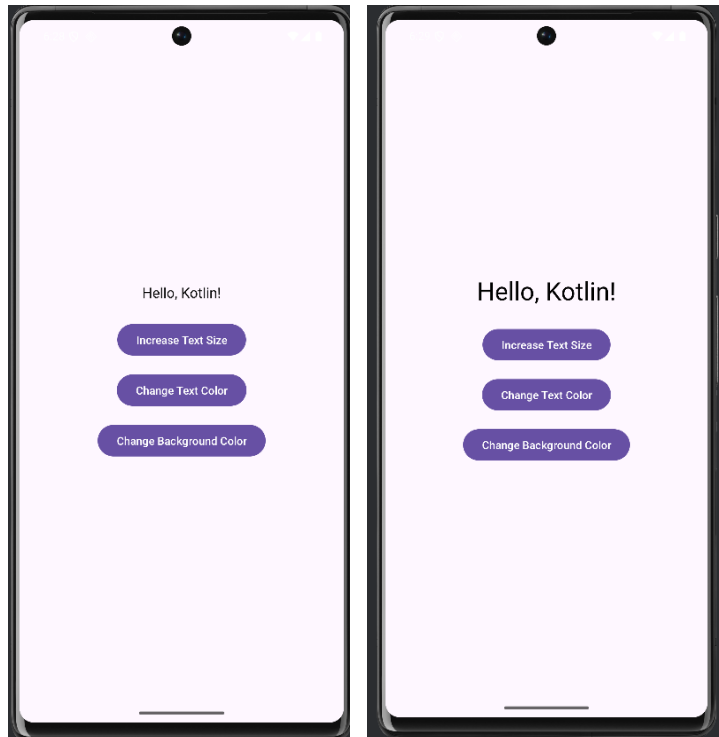
        btnIncreaseSize.setOnClickListener {
            currentTextSize += 2f
            if (currentTextSize > 40f) currentTextSize = 18f // Reset if too big
            sampleText.textSize = currentTextSize
        }

        btnChangeTextColor.setOnClickListener {
            sampleText.setTextColor(getRandomColor())
        }

        btnChangeBgColor.setOnClickListener {
            rootLayout.setBackgroundColor(getRandomColor())
        }
    }
}
```

```
    }  
}  
  
private fun getRandomColor(): Int {  
    val rnd = Random  
    return Color.rgb(rnd.nextInt(256), rnd.nextInt(256), rnd.nextInt(256))  
}  
}
```

Output



Result:

The GUI components experiment has been successfully completed